University of Windsor – Master of Management
BSMM.8730 Data Acquisition and Management | Dr. Andreas S. Maniatis | Fall 2026

D.3.2 Individual Contribution to Project Progress

Charging the Gap: EV Adoption vs. Public Charging Infrastructure in Ontario – Team #8

Nidhi Chetankumar Tandel (110173229) – Project Manager / Scrum Master and Data Engineer |
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1. Role in the Team

In our three-member team I act as Project Manager and Scrum Master and as the team's Data Engineer. As Project Manager I am responsible for planning the sprints, maintaining the Jira backlog, coordinating deadlines and keeping the deliverables consistent with each other and with the instructor's feedback. As Data Engineer I am responsible for the data acquisition and storage layers of the application: retrieving the source data, building the Python pipeline that cleans and integrates it, and designing and loading the MySQL database that the dashboard will use.

2. Work Breakdown

Table 1 summarizes the tasks assigned to me in Jira and their status at the midterm point. The full history of these tasks is available on the Jira board [Jira board link] and the related commits are visible in the GitHub repository [GitHub repository link].

ID	Task assigned to me	Status	Notes / evidence
T1	Problem definition and D.1 pre-proposal (with team)	Completed	Submitted Sep 27
T2	D.1 update: data types, web scraping, MySQL	Completed	Revised after instructor feedback, Oct 2
T3	Jira Scrum board and GitHub repository set-up	Completed	Backlog, six sprints, folder structure, READMEs
T4	Charging-station acquisition from the NLR/NRCan API	Completed	4,608 Ontario stations in JSON
T5–T6	EV-by-FSA, StatCan and boundary file downloads	Completed	Sources and dates logged in data README
T7	Web scraper for the Ontario Data Catalogue links	Completed	Requests + BeautifulSoup
T9–T10	Cleaning, integration and metric calculation notebook	Completed	With validation rules defined by Temitope
T11–T12	MySQL schema, SQL dump, import and verification	Completed	ev_charging_gap with 5 tables and a view
T13	D.4.1 User Requirements and Analysis document (lead)	Completed	Submitted Oct 23
T14	D.4.2 Analysis and Design document (coordination)	Completed	Submitted Oct 30
T17	Read-only MySQL user and connection guide for Power BI	Remaining	Sprint 4, supports Temitope's dashboard
T20	Code comments, README, deployment document (D.4.4)	Remaining	Due Dec 1

3. Contribution Analysis

3.1 Project management

At the start of the project I organized the work into a Scrum structure with two-week sprints aligned to the phases of the Data Analytics Lifecycle, and loaded the backlog and sprints into Jira so that every deliverable had an owner and a date. When the instructor asked us to strengthen the data acquisition phase, I translated the feedback into new backlog items for the semi-structured API data, the web scrapers and the MySQL import, and updated the proposal accordingly. When Shiv rejoined the team in October, I briefed him on the work completed so far and reassigned the news-

release scraper and the projection module to him so that the workload is shared across all three members.

3.2 Data acquisition and preparation

I obtained an API key and retrieved the Ontario charging-station data from the NLR Alternative Fuel Stations API, which serves the Natural Resources Canada station locator, and confirmed that the 4,608 records returned matched the total reported by the API. I downloaded every quarterly EV-by-FSA file from the Ontario Data Catalogue, the Statistics Canada registration and population tables and the FSA boundary file, and documented the source, link and download date of each file in the repository. I then built the Python notebook in Google Colab that flattens the nested JSON, converts missing port counts to zero, standardizes postal codes and city names, derives the FSA of each station, appends all EV quarters into one table and calculates EVs per public port, ports per 1,000 residents and the gap score. I also wrote the scraper that collects the quarterly file links from the catalogue page, which satisfies the web-scraping requirement for the structured data.

3.3 Database design and implementation

I designed the MySQL schema with five related tables and primary and foreign keys, together with the v_fsa_gap view that joins EV registrations, charging ports and population for every FSA and quarter. The notebook generates a self-contained SQL dump, which I imported into MySQL Workbench through the Data Import function, and I wrote verification queries that return the row counts, the most underserved FSAs, the Level 2 and DC fast port split by region and the funding announcements. These results confirm that the data foundation of the application is working and ready for the Power BI dashboard.

4. Challenges and Lessons Learned

The main challenge was that the sources do not line up: stations are point locations with postal codes, EVs are counted by FSA, and the data is published quarterly in one source and as a current snapshot in another. Deriving the FSA from postal codes and building one long quarterly table solved most of this, while stations without a valid postal code will be handled by a spatial join in the next sprint. A second challenge was technical: the domain of the station API changed during the project and the news-release search page did not parse as expected, which taught me to store every raw file in GitHub, log each acquisition step and always plan a fallback. Finally, keeping the proposal, the requirements and the design documents consistent showed me the value of writing measurable acceptance criteria early, because they now guide what the team builds.

5. Next Steps

In the next sprint I will create a read-only MySQL user and a connection guide so that Temitope can connect Power BI to the database, support Shiv in completing the news-release scraper and loading the funding announcements, add the spatial join for stations without a valid postal code, and extend the in-code comments and the README in preparation for the D.4.3 application, the D.4.4 deployment document and the final report due on December 1.

Links

GitHub repository: [GitHub repository link] Jira board: [Jira board link]